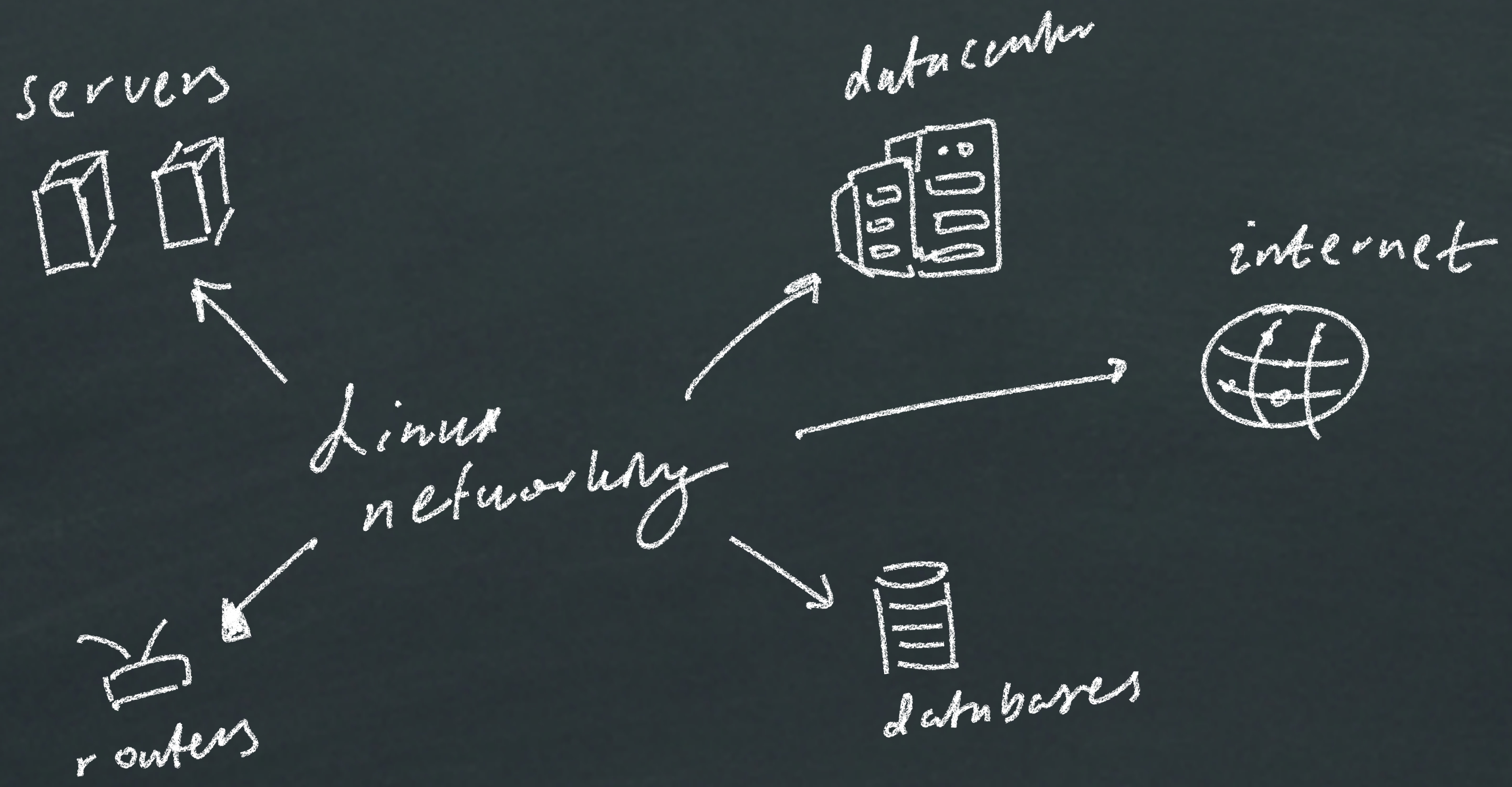
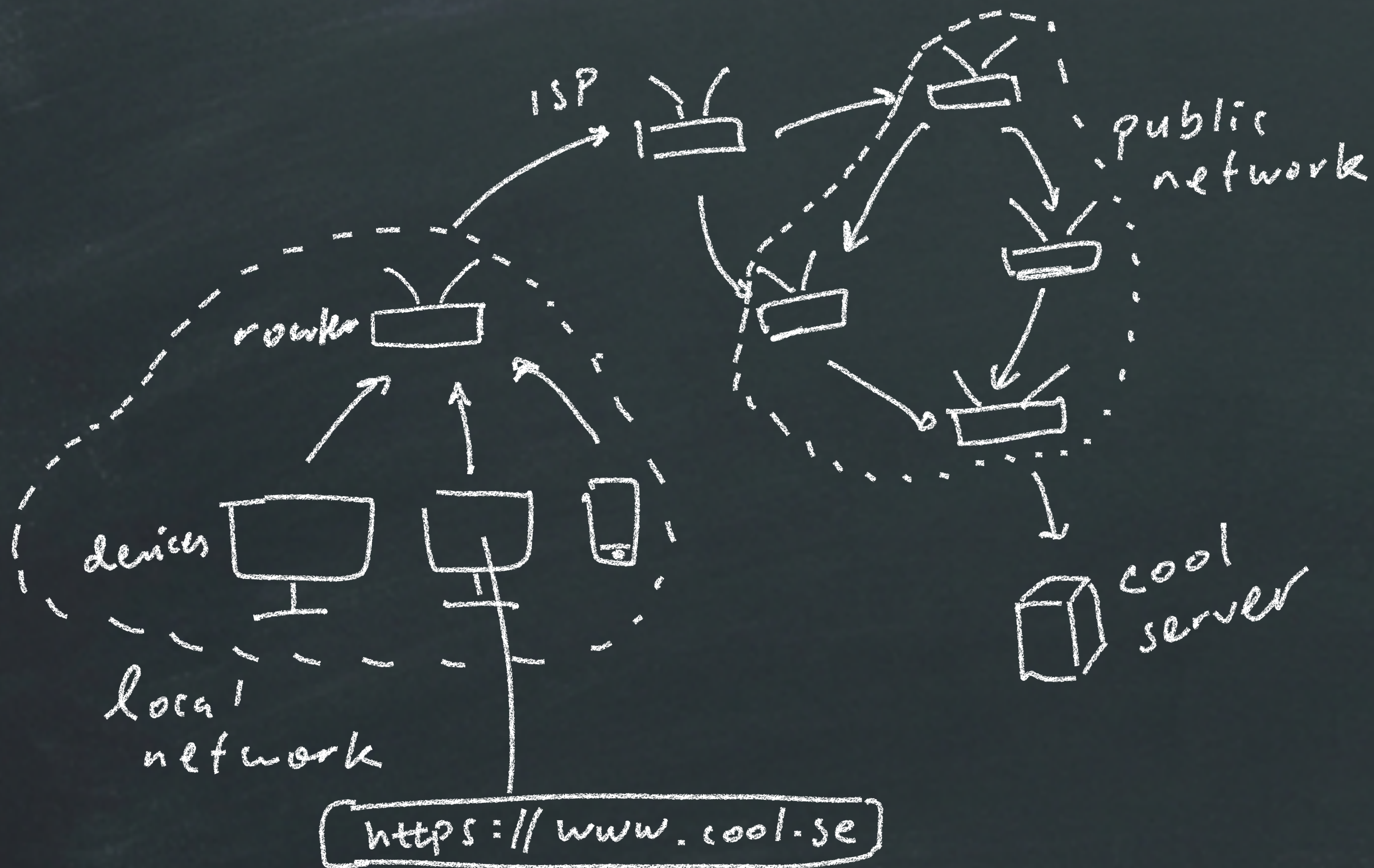
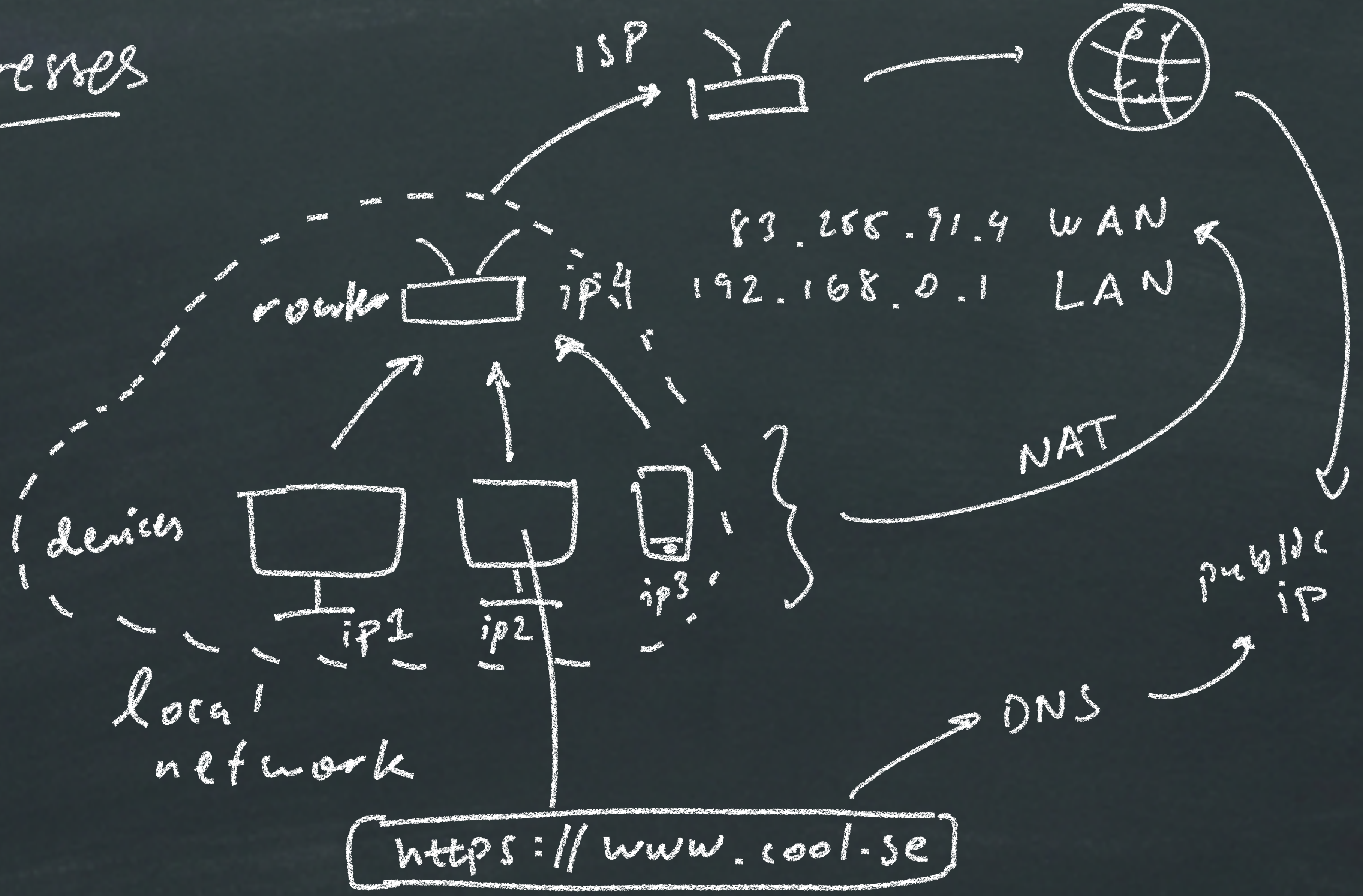
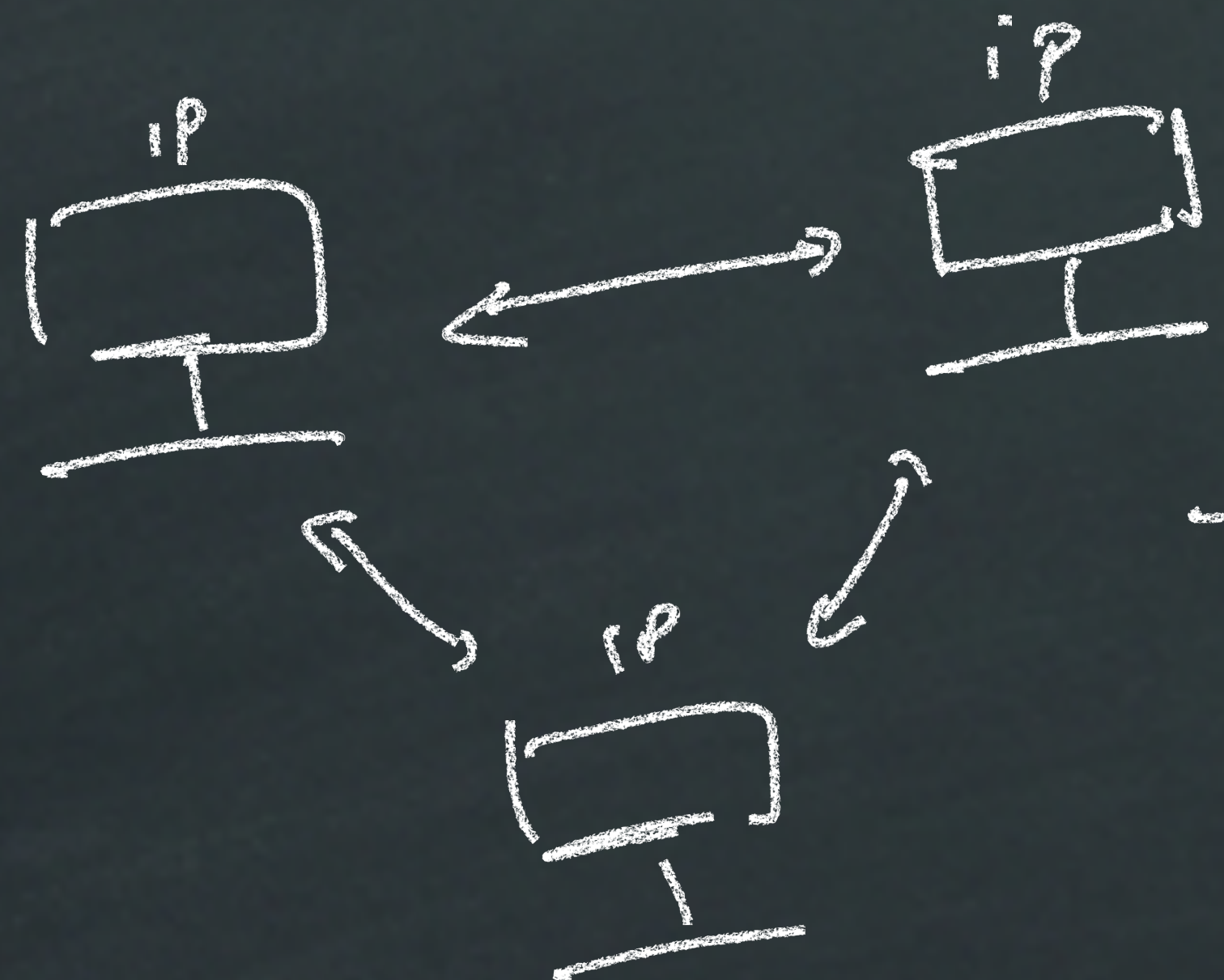






IP addresses





IP address

IPv4

IPv6

$\sim 3.4 \cdot 10^{38}$

ex
192.168.1.1

ex
2001:0db8:85a3:
0000:0000:8a2e:
0370:7334

max
 $\sim 4.3 \cdot 10^9$
devices

IPv4
+
NAT

Calculation

IPv4

xxx.xxx.xxx.xxx

0-255

binary: 8 bytes / group

$$\Rightarrow 2^8 = 256$$

combinations:

$$2^8 \cdot 2^8 \cdot 2^8 \cdot 2^8 = (2^8)^4 = \\ = 2^{8 \cdot 4} = \underline{\underline{2^{32}}}$$

IPv6

FFFF:FFFF:FFFF:FFFF:

FFFF:FFFF:FFFF:FFFF

hexadecimal: 0-9, A-F
16

$$\# \text{ combinations: } (2^{16})^8 = \underline{\underline{2^{128}}}$$